

Treat the following exercises:

Exercise 1 (4 pts)

Convert the following DTD into XML schema:

```
<!ELEMENT ItemInfo (ItemName?, SKU, Quantity, Weight, Price+, Note?, Divider?)>
<!ATTLIST ItemInfo isbn ID #REQUIRED>
<!ELEMENT ItemName (#PCDATA)>
<!ELEMENT SKU (#PCDATA)>
<!ELEMENT Quantity (#PCDATA)>
<!ELEMENT Weight (#PCDATA)>
<!ATTLIST Weight unitOfMeasure (lb | oz) "oz"
                measurementType CDATA #FIXED "US" >
<!ELEMENT Price (#PCDATA)>
<!ELEMENT Note (#PCDATA | Link | Emph)* >
<!ELEMENT Emph (#PCDATA)>
<!ELEMENT Link (#PCDATA)>
<!ELEMENT Divider EMPTY >
```

Exercise 2 (5 pts)

Write the XSL file corresponding to the following XML file and that gives the output below:

```
<?xml version = "1.0"?>
  <?xml:stylesheet type = "text/xsl" href = "games.xsl"?>
  <sports>
    <game id = "783">
      <name>Cricket</name>
      <paragraph>
        Popular among commonwealth nations.
      </paragraph>
    </game>
    <game id = "239">
      <name>Baseball</name>
      <paragraph>
        Popular in America.
      </paragraph>
    </game>
    <game id = "418">
      <name>Soccer (Football)</name>
      <paragraph>
        Popular sport in the world.
      </paragraph>
    </game>
  </sports>
```

PROGRAM OUTPUT



ID	Sport	Information
783	Cricket	Popular among commonwealth nations.
239	Baseball	Popular in America.
418	Soccer (Football)	Popular sport in the world.

Exercise 3 (5 pts)

Write a DTD for the following XML document. Then, change the appearance order of the two elements of the source document. Validate the document and make changes in the DTD in order to obtain a valid document.

```
<?xml version="1.0"?>
<drugs>
<drug>
<name>aspirin</name>
<content> 0.4 g. acetisalicil acid, 0.24 g.
ascorbine acid </content>
<therapy>analgetic </therapy>
<available>yes</available>
<analogon>andol</analogon>
<contraindication>ulcer</contraindication>
</drug>
</drugs>
```

Exercise 4 (6 pts)

Write the XML file as well as the DTD file that correspond to the following schema:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <!-- definition of simple type elements -->
  <xs:element name="name" type="xs:string"/>
  <xs:element name="name" type="xs:string"/>
  <xs:element name="title" type="xs:string"/>
  <xs:element name="author" type="xs:string"/>
  <!-- definition of attributes -->
  <xs:attribute name="isbn" type="xs:string"/>
  <!-- definition of complex type elements -->
  <xs:element name="topic">
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="name"/>
        <xs:element ref="book"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
  <xs:element name="book">
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="title"/>
        <xs:element ref="author"/>
      </xs:sequence>
      <xs:attribute ref="isbn"/>
    </xs:complexType>
  </xs:element>
  <xs:element name="bookstore">
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="name"/>
        <xs:element ref="topic"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```